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Roll No

AU-704-GS
B.E. VII Semester
Examination, December 2020
Grading System (GS)
Combustion and Heat Transfer
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain Combustion stages in SI and CI engines with the help of P-Q diagram.
2. a) Write Fourier equation.
b) What is Thermal Diffusivity?
3. Derive mathematical model for heat dissipation from an infinitely long fin tip.
4. a) With the help of schematic diagram explain burette type volumetric flow meter.
b) Explain infrared spectrographic method used for measurement of flame temperature.
5. Discuss following dimensionless numbers as connected to convection heat transfer
 - a) Nusselt number
 - b) Reynold number
 - c) Prandtl number

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6. Explain fin performance. How to heat dissipation from and infinitely long fin.
7. Comparison between Heat conduction through Composite wall, cylindrical wall, Multilayer cylindrical wall.
8. a) Explain temperature distribution and thermal stresses in Piston and cylinder liners.
b) Briefly explain the need, characteristics and types of cooling systems.
